THURSDAY 5 NOVEMBER 2026 · 09:00–10:00 · GROUP 07

Introduction to Pure Mathematics

Cardinality, the Pigeonhole Principle and permutations · PRIORY RD CMPLX F BLK 3F11

Try a first step before each board discussion. Write definitions and reasons, not just final answers. Question labels match the board.

START HERE · 01–04 MINUTES

Key ideas: cardinality and permutations

Introduction to Pure Mathematics · Week 7 · Section 6.5 and Section 7.1 · Keep this page beside your working.

1 · Comparing sizes with functions

For a finite set, $|A|$ is the number of *distinct* elements. A set inside a set counts as one element: $\{1, \{1, 2\}\}$ has two elements.

Same size. $|A| = |B|$ means there is a **bijection** $A \to B$.
At most. $|A| \leqslant |B|$ means there is an **injection** $A \to B$ (Definition 6.31; for finite sets this agrees with counting, Proposition 6.30). $|A| < |B|$ means $|A| \leqslant |B|$ and $|A| \neq |B|$.

If $A \subseteq B$, the inclusion $a \mapsto a$ is injective, so $|A| \leqslant |B|$. For infinite sets a proper subset can have the same size: $n \mapsto 2n$ is a bijection $\mathbb{Z} \to 2\mathbb{Z}$.

Schröder–Bernstein (Theorem 6.32). If $|A| \leqslant |B|$ and $|B| \leqslant |A|$, then $|A| = |B|$. You may use it; you do not need its proof.

2 · Counting rules for finite sets

$$A \cap B = \varnothing \implies |A \cup B| = |A| + |B| \quad (\text{Prop. } 6.33), \qquad |A \cup B| = |A| + |B| - |A \cap B| \quad (\text{Cor. } 6.34).$$

Pigeonhole Principle. If $|A| > |B|$, then no function $f : A \to B$ is injective: some $a_1 \neq a_2$ have $f(a_1) = f(a_2)$. It is the contrapositive of "injection $\Rightarrow |A| \leqslant |B|$". To use it, say clearly *what the pigeons are, what the holes are, and what the function is*.

3 · Permutations (Section 7.1)

A **permutation** of X is a bijection $X \to X$ (Definition 7.1). The six permutations of $\{1, 2, 3\}$, with the lecture labels:

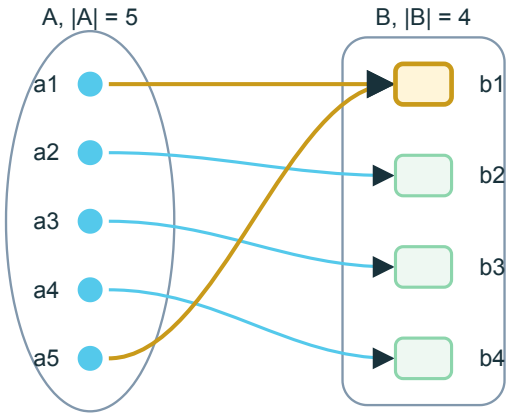
x	e	f	g	h	i	j
$1 \mapsto$	1	1	2	3	2	3
$2 \mapsto$	2	3	1	1	3	2
$3 \mapsto$	3	2	3	2	1	1

Composition is read right to left: $fg = f \circ g$ means "do g first, then f". For example $gf = i$ but $fg = h$, so the order matters. Composites of bijections are bijections (Corollary 6.19), and every permutation has an inverse permutation.

Proof checklist. Same size → write down a bijection and prove it is one (or give injections both ways and quote Theorem 6.32). Pigeonhole → define the function and compare sizes. Finite counting → split into *disjoint* pieces first.

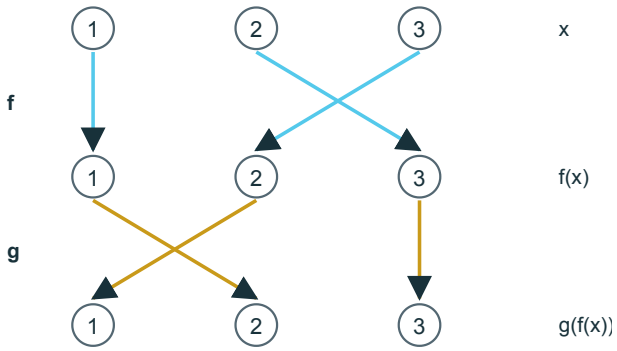
VISUAL · USE DURING P3–P4 AND P8

Pigeonhole and composing permutations



More inputs than boxes ⇒ no injection (Pigeonhole)

5 > 4: whatever the arrows are, two inputs must share a box. Contrapositive of Prop 6.30.



gf = g∘f = i (right-to-left: f first)

gf = i, but fg = h: the order of composition matters.

Tutorial questions

P1 · Counting carefully: what exactly is an element?

Pure PS6 A1(a)–(d)

A1. Calculate the cardinality of the following finite sets:

(a) $A = \{1, 2, \{1, 2\}, 12\}$

(b) $B = \{x \in \mathbb{Z}_+ : 1 \leqslant x \leqslant 30,\ \gcd(x, 30) = 1\}$

(c) $C = \mathrm{im}(f)$, where $f\colon X \to \mathbb{Z}$ with $f(x) = x^2$ and $X = \{x \in \mathbb{Z} : -5 \leqslant x \leqslant 3\}$

(d) $D = \bigcap_{n \in \mathbb{Z}_+} (-1/n,\ 2/n)$, where $(-1/n,\ 2/n) = \{x \in \mathbb{R} : -1/n < x < 2/n\}$

P2 · Finite or infinite? (oral round)

Pure PS6 A2(a)–(d)

A2. For each of the following, determine if the given set is finite:

(a) $A = \{x \in \mathbb{Z}_+ : \gcd(x, 30) = 1\}$

(b) $B = \{x \in \mathbb{Q} : \exists\, y \in \mathbb{Q},\ xy = 2\}$

(c) $C = \mathrm{im}(f) \cap \mathbb{Z}$, where $f\colon \mathbb{R} \to \mathbb{R}$ is the sine function $f(x) = \sin(x)$

(d) $D = \bigcup_{n \in \mathbb{Z}_+} (-1/n,\ 2/n)$, where $(-1/n,\ 2/n) = \{x \in \mathbb{R} : -1/n < x < 2/n\}$

P3 · The Pigeonhole Principle and the birthday-month party

Pure PS6 A4(a)(b) + A5

A4. Let A and B be finite sets with $|A| = m$ and $|B| = n$.

(a) Prove that there is no surjective function $f\colon A \to B$ if $m < n$.

(b) Prove that there is no injective function $f\colon A \to B$ if $m > n$.

A5. What is the smallest number of people you need to invite to a party in order to guarantee that at least two of your invited guests have birthdays in the same month of the year?

P4 · Two people with the same number of friends

Pure PS6 B3

B3. Let $n \geqslant 2$ be an integer. Show that in any group of n people, two of them will have the same number of friends within this group. (You may assume that if person A is friends with person B, then B is also friends with A.)

P5 · Transitivity of $\leqslant$, and of strict inequality

Pure PS6 A6 + B1

Recall (Definition 6.31). For any sets A, B we write $|A| \leqslant |B|$ if there exists an injective function $A \to B$, and $|A| < |B|$ if $|A| \leqslant |B|$ and $|A| \neq |B|$. *Schröder–Bernstein Theorem (Theorem 6.32):* if $|A| \leqslant |B|$ and $|B| \leqslant |A|$, then $|A| = |B|$.

A6. Let A, B and C be sets such that $|A| \leqslant |B|$ and $|B| \leqslant |C|$. Prove that $|A| \leqslant |C|$.

B1. Let A, B and C be sets such that $|A| < |B|$ and $|B| < |C|$. Prove that $|A| < |C|$.

P6 · Inclusion–exclusion from disjoint pieces

Pure PS6 A7 (= Corollary 6.34)

A7. Let A and B be non-empty finite sets. Prove that $|A \cup B| = |A| + |B| - |A \cap B|$.

[You may use Proposition 6.33: if A and B are disjoint finite subsets of a set X, then $|A \cup B| = |A| + |B|$.]

P7 · A bijection from (0,1) onto the whole real line

Pure PS6 B4

B4. Recall that in Question B2 on Problem Sheet 5, we showed that the function

$$g\colon \mathbb{R} \to (-1,1), \qquad g(x) = \frac{x}{1+|x|}$$

is bijective. Using this, prove that there exists a bijection $\varphi\colon (0,1) \to \mathbb{R}$.

[Hint. First construct a bijection $f\colon (0,1) \to (-1,1)$.] (Here $(a,b) = \{x \in \mathbb{R} : a < x < b\}$.)

P8 · Composing permutations of {1,2,3}: order matters

Lecture 7.1.1 (3 Nov) + Pure PS6 B5 (partial)

In lectures the six permutations (bijections $X \to X$) of $X = \{1, 2, 3\}$ were labelled:

- e: $e(1) = 1,\ e(2) = 2,\ e(3) = 3$ (the identity)
- f: $f(1) = 1,\ f(2) = 3,\ f(3) = 2$
- g: $g(1) = 2,\ g(2) = 1,\ g(3) = 3$
- h: $h(1) = 3,\ h(2) = 1,\ h(3) = 2$
- i: $i(1) = 2,\ i(2) = 3,\ i(3) = 1$
- j: $j(1) = 3,\ j(2) = 2,\ j(3) = 1$

We write xy for $x \circ y$, so $(xy)(k) = x(y(k))$: **do y first, then x** (right to left).

(a) Compute gf and fg, and identify each as one of e, f, g, h, i, j. Is $gf = fg$?

(b) Compute $f^2 = ff$ and $i^2 = ii$.

(c) Find the inverse permutation i^{-1}.

(d) In the table whose entry in row x and column y is xy, fill in the row of f: $fe,\ ff,\ fg,\ fh,\ fi,\ fj$.

P9 · Symmetries of a square as permutations of its vertices

Pure PS6 A8

A8. Describe all the symmetries of a square, both geometrically (i.e. in terms of rotations and reflections) and in terms of adjacency preserving permutations of the set of vertices of the square.

Label the vertices as on the sheet (1 top right, 2 top left, 3 bottom left, 4 bottom right):

2	1
3	4

so the edges are $\{1,2\}, \{2,3\}, \{3,4\}, \{4,1\}$, and $\{1,3\}, \{2,4\}$ are diagonals. A permutation σ of $\{1,2,3,4\}$ is *adjacency preserving* if whenever vertices a and b are joined by an edge, $\sigma(a)$ and $\sigma(b)$ are also joined by an edge.

(Follow-up: why is the permutation that swaps 1 and 2 and fixes 3 and 4 not a symmetry?)

After-class practice

Full solutions to these are on Blackboard with Problem Sheet 6.

R-A3 · Reserve: size of a Cartesian product

Pure PS6 A3

A3. Let A and B be non-empty finite sets. Prove that $|A \times B| = |A||B|$, where $A \times B$ is the Cartesian product of A and B.

R-B2 · Reserve: union of n pairwise disjoint sets (induction)

Pure PS6 B2

B2. Let $n \geqslant 2$ be an integer and let $A_1, \ldots, A_n$ be non-empty finite sets that are pairwise disjoint, so $A_i \cap A_j = \emptyset$ for all $i, j \in \{1, \ldots, n\}$ with $i \neq j$. Use induction on n to prove that

$$|A_1 \cup \cdots \cup A_n| = |A_1| + \cdots + |A_n|.$$

[You may use Proposition 6.33: if A and B are disjoint finite subsets of a set X, then $|A \cup B| = |A| + |B|$.]

Adapted from Introduction to Pure Mathematics Problem Sheet 6 (University of Bristol, 2026/27) for tutorial use. No solutions are included.